

SIAGA Data Dictionary

This document lists every dataset used, the harmonized panel schema, and the engineered features that feed the model. It is the reference for reproducing or auditing the pipeline.

1. Source datasets

All files were supplied by the ASEAN Public Health Task Force (ASEAN Statistical Yearbook lineage) and cover the same ten member states: Brunei Darussalam, Cambodia, Indonesia, Lao PDR, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam. Years are mostly 2004 to 2014 unless noted.

Source file	Indicator	Unit	Role in SIAGA	Coverage (2004-2014)
life_expentancy_rate.csv	Life expectancy at birth (by sex)	years	Target (averaged M/F)	89%
goverment_expence_in_health.csv	Capital health expenditure	million current US\$	Feature (per-capita, logged)	91%
physicans_density.csv	Physicians	per 1,000 population	Feature	55%
nurses _ midwife_density.csv	Nurses and midwives	per 1,000 population	Feature	54%
pharmaceutical_worker_density.csv	Pharmaceutical personnel	per 1,000 population	Feature	41%
immunization_DPT.csv	DPT immunization, 1-year-olds	percent	Feature	83%
immunization_measless.csv	Measles immunization, 1-year-olds	percent	Feature	99%
TB_Prevalence.csv	Tuberculosis prevalence	per 100,000	Feature + forecast target	68%
HIV_Prevalence.csv	HIV prevalence	percent	Feature (interpolated)	13% raw
undernourished_population.csv	Undernourished population	percent	Feature	90%
crude_birth_ratio.csv	Crude birth rate	per 1,000	Feature	88%
malaria_prevalence.csv	Malaria cases	per 100,000	Forecast target (bonus)	51%
maternal_mortality_rate.csv	Maternal mortality	per 100,000 live births	Equity layer (SDG 10)	50%
Infant_mortality_rate.csv	Infant mortality	per 1,000 live births	Equity layer, excluded from features (leakage)	86%

under_5_mortality_rate.csv	Under-5 mortality	per 1,000 live births	Equity layer, excluded from features (leakage)	72%
crude_death_ratio.csv	Crude death rate	per 1,000	Context, excluded from features (leakage)	89%
underweight_children.csv	Underweight children under 5	percent	Context / burden narrative	24%
death_by_HIV_AIDS.csv	Deaths due to HIV/AIDS	count	Burden narrative	sparse
crude_birth_ratio.xlsx	(duplicate of crude_birth_ratio.csv)	-	Excluded (verified duplicate)	-
SDG3_summary.csv	Source inventory and links	-	Reference metadata, not modeled	-

External source used for validation and per-capita normalization:

Source	Indicators	Use
World Bank Open Data API v2	Life expectancy, physicians, nurses, immunization, undernourishment, TB incidence, population, crude birth rate, GDP per capita (NY.GDP.PCAP.CD), basic sanitation (SH.STA.BASS.ZS)	Cross-source cleaning check; out-of-source model validation (2015-2019); population for per-capita spend; GDP and sanitation as contextual model features

2. Harmonized panel schema

`data/clean/asean_panel.csv`, one row per country-year.

Column	Type	Description
country	string	Canonical country name
iso3	string	ISO 3166-1 alpha-3 code
year	int	Calendar year
life_expectancy	float	Mean of male and female life expectancy at birth (years)
life_expectancy_male / _female	float	Sex-specific life expectancy
govt_health_capex_musd	float	Capital health expenditure (million US\$)
physicians_per_1000	float	Physician density
nurses_midwives_per_1000	float	Nurse and midwife density
pharma_workers_per_1000	float	Pharmaceutical personnel density
immunization_dpt / immunization_measles	float	Immunization coverage (percent)
tb_prevalence	float	TB per 100,000

hiv_prevalence	float	HIV prevalence (percent)
undernourished_pct	float	Undernourished population (percent)
crude_birth_rate / crude_death_rate	float	Per 1,000 population
infant_mortality / under5_mortality / maternal_mortality	float	Mortality indicators (equity layer)
malaria_cases	float	Malaria per 100,000
underweight_children_pct	float	Underweight children under 5 (percent)

3. Model features and engineering

The model uses fourteen features. Mortality indicators are deliberately excluded to prevent target leakage (life expectancy is computed from mortality tables).

Feature	Source	Engineering	Monotonic prior	Kind
physicians_per_1000	physicians density	interior gap interpolation	+ (more never lowers LE)	actionable
nurses_midwives_per_1000	nurse density	interior gap interpolation	+	actionable
pharma_workers_per_1000	pharma density	interior gap interpolation	+	actionable
immunization_dpt	DPT file	none	+	actionable
immunization_measles	measles file	none	+	actionable
tb_prevalence	TB file	outlier repair, interpolation	- (more never raises LE)	actionable
hiv_prevalence	HIV file	snapshot interpolation (2000/2005/2010/2016 to annual)	-	actionable
undernourished_pct	undernourishment file	Singapore floor set to 2.5 (FAO "<2.5%")	-	actionable
crude_birth_rate	crude birth file	none	0 (unconstrained)	context
log_capex_per_capita	spend + WB population	$\log(1 + \text{spend} \times 1\text{e}6 / \text{population})$	+	actionable
log_gdp_per_capita	World Bank GDP per capita	$\log(1 + \text{gdp per capita})$	+	context
sanitation_pct	World Bank basic sanitation	interior gap interpolation	+	actionable
year	index	secular trend term	0 (unconstrained)	context

The `kind` column marks which features the policy simulator exposes as sliders (actionable) versus holds constant as controls (context). GDP per capita and sanitation access are established life-expectancy determinants; adding them lifted the out-of-source MAE from 2.65 to 2.08 years, and SHAP ranks sanitation as the top driver.

Engineering rationale

- **log_capex_per_capita:** absolute million-US\$ spend is not comparable across countries of vastly different population, so it is converted to spend per person and log-transformed to reflect diminishing returns.
- **hiv_prevalence:** reported only at four snapshots; interpolated because population prevalence changes slowly and smoothly.
- **Interior interpolation only:** gaps between observed years are filled within each country; series edges and the target are never filled.
- **Monotonic priors:** enforced in the gradient-boosting model and in the API's linear surrogate, guaranteeing medically sensible predictions.

4. Derived and serving artifacts

File	Contents
data/clean/asean_panel.csv	Harmonized panel
data/clean/modeling_frame.csv	Feature matrix + target used for training
data/clean/models.joblib	Fitted gradient-boosting model + feature list
data/clean/linear_surrogate.json	Sign-constrained linear edge model for the API
data/clean/shap_importance.json	SHAP driver ranking (years of life expectancy)
data/clean/disease_forecasts.csv	5-year TB and malaria forecasts with intervals
data/clean/metrics.json	Validation metrics (R2, RMSE, MAE)
data/clean/worldbank.csv	World Bank indicators for validation
data/clean/quirks_log.md	Every data-quality issue found and the action taken